

The empirical recurrence for OEIS A261550, proved by transfer matrix

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Abstract

OEIS A261550 counts $(n+2) \times 5$ arrays over $\{0, \dots, 1\}$ subject to a condition imposed on every 3×3 subblock, and it carries an empirical recurrence of order 23 contributed by R. H. Hardin. It is true, and it is not an empirical matter. A 3×3 subblock spans three consecutive rows, so the state of a transfer matrix is a PAIR of consecutive rows and appending one more row is a step; an array is then a walk, and $a(n)$ is a walk count on $S = 1024$ vertices. Such a sequence is C -finite, so whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A261550 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000101 or 00001011.”. It has offset 1 and begins

170, 269, 516, 754, 1210, 2187, 3436, 5368, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 9*a(n-6) + 4*a(n-7) + 14*a(n-9) + 12*a(n-10) + 2*a(n-11) + 9*a(n-12) + 4*a(n-13) + 12*a(n-14) - 16*a(n-16) + 24*a(n-17) - 4*a(n-18) - 24*a(n-19) + 22*a(n-20) - 16*a(n-22) + 12*a(n-23)$ for $n > 30$

As of the “Last modified” line on the live entry (Aug 24 2015 10:57:45, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Three consecutive lines

Write an admissible array as a sequence of rows $u_1, u_2, \dots$, each an element of $\{0, \dots, 1\}^5$. A 3×3 subblock occupies three consecutive rows and three consecutive columns, so with r, s, t three consecutive rows the blocks it contributes are

$$g = \begin{pmatrix} r_j & r_{j+1} & r_{j+2} \\ s_j & s_{j+1} & s_{j+2} \\ t_j & t_{j+1} & t_{j+2} \end{pmatrix}$$

for each admissible index, and the entry’s condition is the requirement that

the clockwise perimeter word of g is a rotation of one of $\{00000001, 00000101, 00001011\}$

holds for every such g . This involves r , s and t only: it is a condition on three consecutive rows and on nothing else.

Take as vertices of a digraph G the ordered PAIRS (r, s) of rows, of which there are $S = (1 + 1)^{2 \cdot 5} = 1024$, and put an edge $(r, s) \rightarrow (s, t)$ exactly when the condition holds for the triple (r, s, t) at every admissible index. An array with L rows is then a walk of length $L - 2$ in G : its first two rows name the starting vertex and each further row is one step. Hence, with M the adjacency matrix of G ,

$$a(n) = \mathbf{1}^\top M^{n-1+1} \mathbf{1},$$

the exponent being fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^{23} - \sum_i c_i t^{23-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n-i)$ equals $\mathbf{1}^\top M^j q(M) \mathbf{1}$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$ for all large j ; and by the Cayley-Hamilton theorem it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+23} - \sum_i c_i M^{j+23-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$, so the numbers $u_j = \mathbf{1}^\top M^j q(M) \mathbf{1}$ satisfy the monic linear recurrence given by the characteristic polynomial of M ; S consecutive zeros therefore force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$a(n) = 9a(n-6) + 4a(n-7) + 14a(n-9) + 12a(n-10) + 2a(n-11) + 9a(n-12) + 4a(n-13) + 12a(n-14) - 16a(n-15)$
for every $n > 30$.

Proof. The vector $w = q(M) \mathbf{1}$ was formed by 23 matrix-vector products in exact integer arithmetic and $\mathbf{1}^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly. Those integers vanish from the index corresponding to $n = 30 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 30$. The bound is exact: at $n = 30$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 30 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts. That check did real work here. The phrase "clockwise perimeter pattern" does not mean that the eight boundary entries, read clockwise from the top left, equal the quoted word; it means the cyclic word they form is that word up to rotation. Read the first way the model returns counts far below the entry's own first term, and only the second reading reproduces the published values. It is the one used.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix, so no rounding or sampling enters anywhere.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.